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Additional Information

A Smart IoT-based Irrigation System Design using AI and Prediction Model

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Abstract:

Implementing intelligent irrigation and adjusting the irrigation system to control the amount of water required for the plant is essential in today's agricultural system. This study focuses on data obtained from sensors measuring (soil temperature and humidity, temperature and humidity of air and light), and data obtained from image processing of plant leaves (RGB). To analyze the data, a regression model in SPSS software and another model by genetic programming in MATLAB 2018 software were implemented. The optimal model was a combined model of sensors and images in genetic programming with higher R^2 and lower standard error of 0.88 and 0.03, respectively. This model was superior to the regression model which had an R^2 and standard error of 0.86 and 0.21, respectively, so this optimal model was selected to adjust the microcontroller for intelligent irrigation. The following year, with the replanting of the crop, the intelligent irrigation system was presented as the superior system with 11% water saving compared to the previous year (irrigation by the user). Also, no changes were observed in the yield and color indicators of the plant at the level of 5%, which indicates the superiority of the intelligent irrigation system and its high accuracy of this system.

1. Introduction

As the human population and demand for freshwater resources increase, frequent droughts and water scarcity are threatening global food security by severely disrupting agricultural production (Lei et al., 2016). The challenge in recent years in this area is to improve the management methods of consumption of agricultural inputs (especially water) and, accordingly, to increase demand productivity, which requires a deep understanding of plant responses to abiotic stresses (Behmann et al., 2014). Among the methods to increase water consumption and create sustainability in agriculture, the implementation of appropriate timing methods for irrigation and early detection of water stress in crops before causing irreversible damage and reduced yield can be named (Osroosh et al., 2015). Recent studies have focused on remote control data as an alternative to traditional measurements of crop stress parameters, as they provide useful information about spatial and temporal changes in crops and soil (Singh and Sharma, 2022; Wang et al., 2021 Guo, 2021).

Precision farming is a set of technologies that help collect and analyze agricultural-related data. These technologies include GPS, sensors, remote sensing, and other electronic communication devices (Sharma et al., 2018). The main goal of precision agriculture is to maximize agricultural production by focusing on optimizing and improving agricultural processes (Tzounis et al., 2017). In fact, by moving within the framework of the principles of sustainable development, precision agriculture pursues goals such as increasing productivity based on reducing environmental damage, managing the consumption of inputs in the right place and time, and reducing costs. It can be a reason for the certainty of using new technologies related to precision agriculture. Precision farming includes a wide range of electronic equipment (sensors), optical instruments (digital and multispectral cameras, hyperspectral, thermal, spectrometers, and other related instruments), the use of the World Wide Web and the Internet of Things, and includes a large part of computer software that all these technologies are used and applied in the production process from land preparation to harvesting and processing of the product (Bagheri, 2019; Khoshamoz et al., 2021).

Our motivation in this work is

This paper is structured as follows

2. Related Work

2-1. Internet of Things and Intelligence Irrigation system

Precision irrigation, which today plays a significant role in precision agriculture, as an advanced concept in agriculture, promises to improve water use efficiency and maintain or increase crop yield. On the other hand, this requires various advanced technologies such as IoT, wireless sensor networks, and cloud computing (a platform for collecting information and performing calculations in IoT technology) (Rahimi Jahangirloo, 1399). The concept of the Internet of Things was first introduced by Professor Ashton at the MIT Auto-ID Center. The concept of object-to-object connection was also coined in 1995 by Bill Gates. IoT technology involves object-to-object, object-to-human, and human-to-human connectivity, intending to build a vast network of sensors such as remote sensing, GPS, laser scanners, and RFID. The Internet of Things has a huge communication infrastructure in which a large number of heterogeneous nodes are active in the network and also a large amount of data is exchanged in this network which is of macro data type (Boursianis et al., 2020; Sheikh et al., 2020). The most important network in this technology is the wireless sensor network (WSN), which consists of one or more wireless nodes that are connected to the sensors, and these nodes are also responsible for collecting data. The IoT often includes different layers such as the physical layer (collecting all sensor data in it), the network layer (data transfer), and the decision layer (data analysis and processing) (Shafi et al., 2019). The Internet of Things incorporates various communication technologies to connect different objects and also moves towards up-to-date wireless communication (Miloudi and Rezeg, 2018).

Agricultural producers can control the amount of irrigation using a mobile phone-based on plant color data and environmental and soil characteristics. These days, the Internet of Things and predictable models make different solutions possible. Therefore, farmers with limited irrigation capacity can store large amounts of water and irrigate at the best possible time. The main purpose of smartening the system is to supply the required amount of water for agricultural products by monitoring soil moisture and temperature, ambient temperature, and color characteristics of the plant. Accurate irrigation with the help of the Internet of Things is an integral part of precision agriculture and greenhouse management and irrigation, and the farmer can monitor the amount of

irrigation and the yield of his products at any time and any place. Singh and Sharma (2022) reported that by measuring soil moisture, temperature, and ambient humidity by sensors and receiving data through a combination of two technologies the Internet of Things and Unmanned Aerial Vehicles (UAVs), the precision and F of this combined technology is 98.77% and 99.49% respectively. Wang et al. (2021) used the multilayer perceptron neural network and the Internet of Things to predict sugarcane yield and reported that the proposed MLP algorithm had a maximum accuracy of 99%, a precision of 95%, and an MAE of 0.04%.

Pincheira et al. (2021) used the combined IoT and blockchain system as a reliable source of data collection to manage water use in precision agriculture. According to the reported results, the proposed system can be implemented with only 6% of the additional energy budget compared to the performance of normal devices equipped with IoT. Patankar et al. (2021) introduced the IoT-based irrigation system that was able to control the amount of water consumed by measuring soil moisture, humidity, and air temperature. Accordingly, the water required for the plant was reported to be 800-600 mm per day at a temperature of 40 to 600 ° C. Sabzi et al. (2021) estimated the nitrogen content of cucumber leaves using partial least squares regression and hyperspectral imaging data (including RGB visible range and effective (optimal) and ineffective hyperspectral wavelengths) with the neural network. The results showed that the mean regression coefficients (R) for synthetic particle swarm optimization (ANN-PSO) in the range of 0.937-0.965, partial least squares regression (PLSR) 0.975-0.997, and one-dimensional deep neural learning networks (NN). Range 0.965-0.985. Therefore, it was reported that among the non-destructive hyperspectral images of leaves, regression models have a significant ability to accurately predict the amount of nitrogen in cucumber plants. Shi et al. (2020) estimated rice nitrogen nutrition with RGB images and machine learning techniques. In this study, were used a digital camera for recording RGB images of rice canopy and three regression methods (simple nonlinear regression, SNR; backpropagation neural network, BPNN; and random forest regression, RF) to estimate rice shoot dry matter (DM), estimates N accumulation (NA) and leaf area index (LAI). Random forest regression (RF) models had the highest accuracy for the data set with average testing prediction accuracy (ATPA) of 80.17, 79.44%, and 81.82% for estimating DM, LAI, and NA, respectively. The highest standard deviation (SD) and interval (5% - 95%) of ATPA were observed in BPNN models, indicating that the BPNN model has the most susceptible to dataset splitting.

Gholizadeh et al. (2020) used Sentinel-2 remote sensors to measure and estimate soil organic carbon with soil color images, from different color spaces including reflectance spectroscopy at selected wavelengths in the visible range (700-400 nm), RGB digital cameras, and visible bands, as well as, they used machine learning (RF) for modeling. The results of RGB digital camera images were obtained with higher R^2 and less error (R^2 = 0.85, RMSEP = 0.11%) compared to spectroscopy (R^2 = 0.78, RMSEP = 0.09%). Meanwhile, Sentinel-2 remote control sensors predicted soil organic carbon with less precision. Therefore, using an RGB digital camera is more efficient and cheaper than the other two methods. Perez-Rodrigues and Gomes-Garcia (2019) performed regression-based processing to classify RGB plant color images. By imaging the pine plant, they concluded that the indices R and G (R = f (G)) have the highest Pearson correlation coefficient (0.914) and the highest R^2 (R^2 = 0.835). They presented the regression model y = -0.0005241 X2 +1.020 X + 40.49. K˘rí˘zov´a et al. (2022) used a one-dimensional computer, a Pi camera, and a simple LED in a three-dimensional printing box. They obtained a prototype called Rasp2SPAD through color images and spectral indices to determine plant chlorophyll (SPAD), which was modeled for plant SPAD by a simple linear model, a generalized linear model, and an artificial neural network. According to the results, using color indices b and r, which were obtained as the most sensitive color indices from the analysis, a simple linear model with mean absolute deviance of 1.85 and R2 = 0.81 was introduced with high accuracy and reduced costs by 5%.

2-2. Genetic Programming

Genetic programming is a complete version of the genetic optimization algorithm, with the difference that genetic programming in each gene Can be considered mathematical operators such as plus, subtraction, multiplication, and division, different functions such as logarithmic and trigonometry functions, as well as the names of input variables, in addition to numbers (values of decision variables). Also, in genetic programming, instead of the usual chromosome structure in the genetic algorithm, we encounter a tree structure of chromosomes. Ashofteh et al. (2019) investigated a bi-objective genetic programming (BO-GP) algorithm to optimize irrigation using reservoir performance. Each operational policy was calculated by BO-GP based on the inflow volume to the reservoir, the storage volume, and the water demand volume that minimize vulnerability and maximize reliability. The results showed that the system vulnerability and reliability are between 16 to 41% and 46 to 78%, respectively. Wen et al. (2022) used genetic

programming (GP) to extract and classify farmland using images compared to five common classification programs including K-Nearest Neighbor (KNN), Decision Tree (DT), Naïve Bayes (NB), Support Vector Machine (SVM), and Random Forest (RF). According to the results, GP can maintain high accuracy with any number of samples compared to other classifiers by using different numbers of training samples. Therefore, the overall precision for KNN, DT, NB, SVM, RF, and GP was 83.3%, 94.4%, 83.3%, 93.1%, 97.2%, and 98.6%, respectively, with higher GP accuracy in the optimal classification.

In this research, to manage the amount of water used for plant irrigation and remote control of the irrigation system by the user (farmer), a system is designed and implemented. This system has developed a model based on environmental factors (light, temperature, and humidity), soil conditions (soil temperature and humidity), and plant color indicators (leaves). Then, it has planned the designed system according to the proposed model, for evaluation, o the irrigation system intelligently performs the irrigation operation. This research is a comprehensive combination of environmental, color, and soil data. Table (1) shows a set of studies in this field that have been conducted focusing on the limited characteristics of factors affecting plant conditions and irrigation.

Table 1. Summary of related work based on modeling, image processing, IoT, regression

	RGB Index	Genetic programming	Internet of Things	Regression	Sensors
Sharma and Singh (2022)			*		*
Wen et al (2022)		*			
Krizova et al (2022)	*				
Pincheira et al (2021)			*		
Wang et al (2021)			*		

Patankar et al (2021)			*		*
Sabzi et al (2021)	*			*	
Gholizadeh et al (2020)	*				*
Shi et al. (2020)	*			*	
Perez-Rodrignes & Gomez-Garcia (2019)	*			*	
Ashofteh et al (2019)		*			
This research	*	*	*	*	*

3. Material and Methods

This study was conducted in Iran, the northeastern region of Khuzestan province located in Khuzestan University of Agricultural Sciences and Natural Resources. In this study, a greenhouse with an area of 35 square meters (7.5 meters) was selected. The cucumber crop was studied in this study, which was sampled from 12 plants and relevant measurements were performed.

3-1. Design and construction of electromechanical part in intelligent irrigation system

In this study, 16 capacitive humidity sensors (model SEN0193, China) with an accuracy of 0.1% were used which measured the amount of soil water volume using the dielectric properties of water. Moisture values were collected by the analog channel of the microcontroller. In addition to the non-contact infrared thermometer module (Model MLX90615, China) with an accuracy of 0.02, 16 soil temperature sensors (DS18B22, Taiwan) with an accuracy of 0.1 degrees Celsius and an optical sensor (MAX44009, China) with an accuracy of 0.01 was also used in the set of sensors, In addition, in the set of sensors, were used from the non-contact infrared thermometer module (Model MLX90615, China) with an accuracy of 0.02, 16 soil temperature sensors (DS18B22, Taiwan) with an accuracy of 0.1 degrees Celsius and an optical sensor (MAX44009, China) with an accuracy of 0.01 that soil temperature and humidity sensors were installed next to each plant to collect data locally. Also, sensors of light, temperature, and humidity of the environment were placed at a height of one meter above the soil surface (Taki et al., 2016). To measure water consumption, a rotating sensor (model YF-S201, China) with an accuracy of 0.1 ml was used along with the Hall effect sensor. Data recording steps were performed by sensors continuously. Figure (1) shows a schematic of an intelligent irrigation system in a greenhouse.

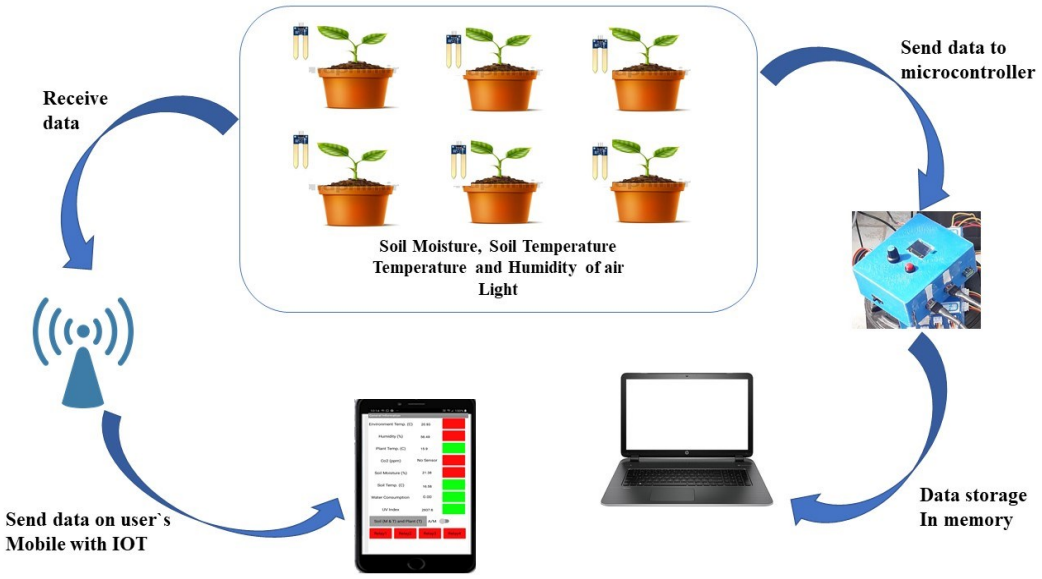
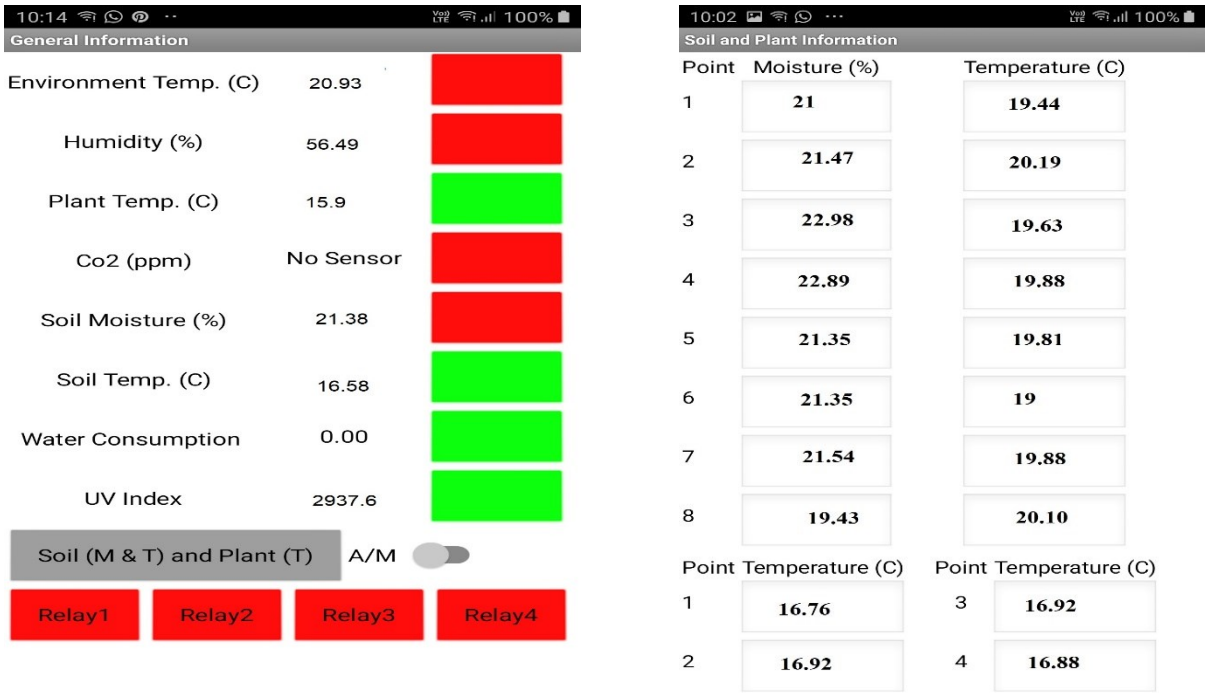


Figure 1. Schematic of storing and transmitting information on a computer and displaying information received from sensors via the Internet on the user's mobile phone

ESP8266 module (ESP-01 model, China) was used for WiFi communication. This low-power module uses a frequency of 2.4 GHz to provide a Wi-Fi connection between 400-500 meters outdoors. All the data obtained from the sensors, in addition to being stored in the device memory, were also visible through the Internet and software written in the Android environment on the user's mobile phone, and the user was able to view the function of each sensor and their data. On his mobile phone (Figure 2 a and b).

The desired application can be displayed by installing it on any smartphone. As shown in Figure 2a, the system can be adjusted in two modes: automatic control (A) and manual control (M). And if the manual control mode is selected, each of the 4 solenoid valves can be turned on or off if necessary, and if the device is set automatically, the solenoid valves are turned on and do irrigation according to the device forecast (which is based on the planning model in it Are). On the right side of the sensor data, there are two colors, green and red. Green indicates the appropriate conditions for that sensor and the plant, but red indicates that the data received (from the sensor) is not in the acceptable range and the user must check.



a **b**

Figure 2. a) Schematic of the user's mobile screen and the status of different parts of the irrigation system; B) Data related to soil temperature and humidity sensors

3-2. Image processing

In this study, to investigate the color characteristics of the plant, imaging was taken every other day after the 2-leaf stage (when the plant height reached 20 to 30 cm and the two main leaves were fully formed) from the leaves at a distance of 20 Cm (perpendicular to the plant). Then, RGB color space was used to examine the images and extract their features. The color characteristics used in this image include the mean red, green, and blue color parameters (Biabi et al., 2019). After extracting the image features, the relevant data were sorted and used for modeling by genetic programming and regression.

3-3. Genetic programing

Understanding the structure of problems, mathematical modeling, and optimizing the coefficients of the developed model can be done simultaneously using GP (Muttil and Lee, 2005). The implementation stages of this model include 1. Training stage and 2. The test stage. In the training phase, the weights are corrected step by step until the minimum prediction error is corrected by the software, which means that first a series of data as an independent variable (soil temperature, light, ambient temperature, ambient humidity, parameters Color r, g, and b) and the dependent variable (soil moisture) were entered into the model as input data. Then, MATLAB 2018 software continued to repeat the calculations by assigning different weights to independent variable data and predicting and calculating the value of the dependent variables and comparing it with the dependent variable data measured by changing the amount of weights step by step to reach the least error. Input data were defined in 3 ways for the software: 1. Color parameters as input data and prediction of soil moisture model based on them, 2. Sensor data (soil temperature, light, ambient temperature, ambient humidity) As input data and prediction of soil moisture model based on them, 3. A combination of sensor data and color parameters as input and prediction of soil moisture model based on a combination of them. In general, in a genetic algorithm, the first sets of chromosomes are considered the initial population and then a new population is created by performing the combination and mutation operations on the chromosomes. Usually, the best case for the initial population, percentage of composition, and mutation in each period is obtained by trial and error (Nadafzadeh et al., 2018). In this study, the values of these parameters were equal to 500, 0.02, and 0.8, respectively. The output of a designed model is displayed in the form of a tree diagram (Figure 3). This tree diagram consists of a set of terminals (problem variables) and nodes (mathematical operators).

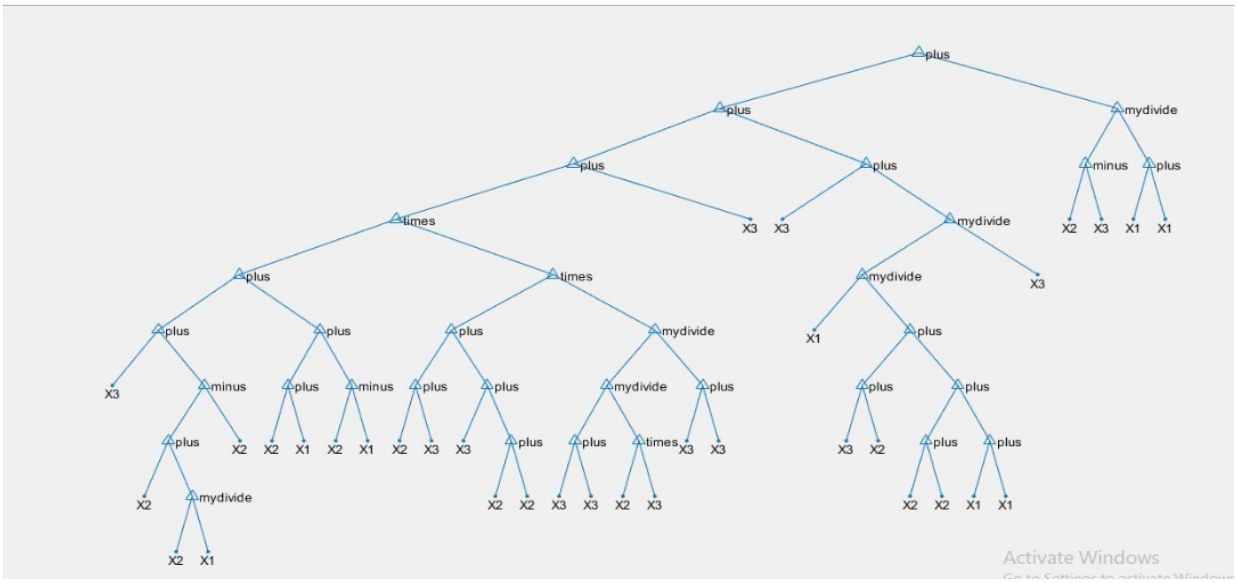


Figure 3. An example of a tree diagram from genetic programming

According to the above figure, x1, x2, x3, x4, x5, x6, and x7 are equivalent to the color parameters r, b, g and soil temperature, ambient temperature, ambient humidity, and light, respectively. In the end, the best models obtained were presented by the graph based on the error rate (RMSE, MSE) and R^2. In this method, at the beginning of the process, a function is not considered and this method can optimize the components of its model components. Genetic programming, unlike genetic algorithms, works on the tree structure of formulas instead of a series of binary digits. To perform modeling operations by GA, first, an initial population randomly is generated to construct the initial tree diagram structure. Then, by crossing the information of two members (in nodes), new information is created that these two members have the role of parent and child in genetics. The one-point intersection method is usually used to select a random point where two members' information is combined. Mutations are determined experimentally based on the calculated fit, which is used for randomly changing the type of mathematical operators in nodes and their values.

In the first stage (training and calibration stage), 70% of the data were used and the remaining 30% of the data were used in the test (validation) stage of the model. For fitting in the software implementation stages, model parameters were selected including various mathematical operators (+, -, ×, /) and measurement criteria (RMSE, MSE, R2). Then the software was run and the standard error (RMSE) was calculated from Equation (1) (Pourgholam-Amiji, et al., 2021).

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n}(Y_i(m) - Y_i(p))^2}{n}} \quad (1)$$

In this formula, Yi (m) is the measured value, Yi (p) is the value predicted by the model, and n is the amount of data.

To present the regression model, regression analysis was performed with SPSS 26.0.0.1 software. Pearson correlation coefficient was performed to investigate the correlation between soil moisture as a dependent factor and independent factors such as color indicators R, G, and B, soil temperature, ambient temperature and humidity, and the amount of light as input factors to the program. Finally, the regression equation was presented with its R^2.

4. Results and Discussion:

4-1. Regression analysis:

Regression analysis was performed after RGB color calculations and sensor data. According to the results of the regression study, the correlation was investigated between soil moisture as a dependent factor, and independent factors such as color indices R, G, and B, soil temperature, ambient temperature, and amount of light as input factors.

Table 2. Correlation between soil moisture and color indices R, G, and B, soil temperature, ambient temperature, ambient humidity, and light level

	S.M	S.T	r	g	b	en.T	LIGHT	HUM

S.M	1							
S.T	-.833**	1						
r	-.804**	.724**	1					
g	-.485**	.409**	.829**	1				
b	-.845**	.760**	.970**	.813**	1			
en.T	-.573**	.486**	.393**	.230**	.438**	1		
LIGHT	-.154	.159	.215	.139	.165	.029	1	
HUM	-.796**	.709**	.677**	.464**	.770**	.465**	.082	1

**. Correlation is significant at the 0.01 level (2-tailed).

b. Cannot be computed because at least one of the variables is constant.

According to Table 2, the variables r, g, and b, soil temperature, temperature, and humidity of the environment have a significant correlation at the level of 5% with each other and with soil moisture. Also, considering the amount of soil moisture, a significant negative correlation was observed between moisture and leaf color indices at the level of 5%. As a significant positive and negative correlation is observed between soil temperature and soil moisture with ambient temperature, respectively, soil temperature increases with increasing ambient temperature, and soil moisture follows a decreasing trend. In soils with higher humidity, the rate of heat transfer and distribution in the soil slows down due to the cooling role of moisture in the soil (Sabziparvar et al., 2010). Also, the relationship between soil temperature and temperature parameters gradually decreases with increasing soil depth, although several factors can lead to changes in soil temperature such as soil profile characteristics, soil texture, thermal conductivity, specific heat, etc. (Baaghideh et al., 2019).

Karthikeyan and Mishra (2021) investigated soil moisture in several layers using a visual machine. The mean ground surface temperature was not significantly correlated with the mean soil moisture during the day but was significantly correlated at night. Also, in another study that examined the relationship between ground surface temperature and soil moisture during the night and early morning, the standard deviation of surface temperature during the day with the average soil moisture had a significant negative correlation, while at night, the correlation was not significant (Zhao et al., 2018). In examining the relationship between soil at different depths and meteorological parameters, it was reported that temperature parameters determine the temperature

status of soil depth, and also this relationship gradually decreases with increasing soil depth. In examining the relationship between soil at different depths and meteorological parameters, it was reported that temperature parameters determine the temperature status of soil depth, and also this relationship gradually decreases with increasing soil depth. Thus, the R^2 between soil temperature and climatic parameters decreased from 0.91 at depths of 5 and 20 cm to 0.81 at depths of 50 cm. Also, there was an inverse relationship between the relative humidity of the air and soil temperature parameters (Baaghideh et al., 2019). On the other hand, the amount of moisture available to the plant is reduced with decreasing soil moisture, so the color indicators are affected and the color of the leaves becomes brighter, Therefore, a significant negative correlation was observed between plant color indices and soil moisture. In a study by Biabi et al. (2019), to investigate the intelligent irrigation system for Lilium, the effects of water stress on the plant were investigated and a significant effect was observed at a 5% level between water stress and plant color indices.

After examining the significant correlation between the measured parameters (temperature and humidity and light of the ambient, soil temperature and soil humidity, plant color indicators) and soil moisture, a model combining was analyzed with the factors mentioned by the Regression analysis method. According to the results, the standardized R^2 is equal to 0.868 and the standard error is equal to 0.21. Also, considering that regression is significant at the 5% level, at least one of the independent variables has a linear relationship with the dependent variable (Table 3). Therefore, the regression model was estimated to calculate the constant value and coefficient of the independent variable in the regression equation (according to Table 4).

Table 3. Regression variance analysis

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	37.614	6	6.269	140.484	.000[b]

	Residual	5.399	121	.045
	Total	43.013	127	

a. Dependent Variable: S.M

b. Predictors: (Constant), HUM, g, en.T, S.T, r, b

Table 4. Calculation of regression coefficients in the regression model

		Unstandardized Coefficients		Standardized Coefficients		
	Model	B	Std. Error	Beta	t	Sig.
1	(Constant)	68.259	10.053		6.790	.000
	S.T	-.472	.177	-.162	-2.660	.009
	r	-4.703	2.290	-.310	-2.053	.042
	g	5.560	.980	.407	5.674	.000
	b	-5.241	1.725	-.571	-3.038	.003
	en.T	-.389	.101	-.145	-3.852	.000
	HUM	-.455	.183	-.153	-2.486	.014

a. Dependent Variable: S.M

The regression equation with R2 = 0.86 is as follows according to Table 4 and the variable and constant coefficients of the desired factors obtained from regression analysis (Formula 2):

$$S.M= 68.259-0.162\ S.T – 0.310\ r + 0.407\ g – 0.571\ b – 0.145\ en.T – 0.153\ HUM \quad (2)$$

In the high regression model, the g index has the highest positive coefficient the model and then the coefficients related to ambient temperature, ambient humidity, soil temperature, r index, and b have the most negative coefficients, respectively. Perez-rodrigues and Gomes-Garcia (2019) performed regression-based processing to segment RGB color plant images. By imaging the pine plant, they concluded that R and G indices (R = f (G)) have the highest Pearson correlation coefficient (0.914) and the highest R2 (0.835) and presented the regression model y = -0.0005241 X2 +1.020 X + 40.49. Krizova et al. (2022) presented a linear model for plant chlorophyll through

color indices and spectral images that color indices (b and r) were the most sensitive color indices of the analysis. A simple linear model with high accuracy (mean absolute deviation of 1.85 and R2 = 0.81) was introduced reduced costs by 5%. On the other hand, the light factor did not have a coefficient in the predicted model and no effect was predicted for it in the model, so the analysis of the results was examined by genetic programming.

4-2. Genetic Programming analysis:

To provide the optimal model for predicting soil moisture, the predicted models are presented by genetic programming (Table 5). Table 5 shows the models predicted by the software based on color parameters r, g, and b (Equations 6-1), data recorded from sensors (Equations 7-12), and a combination of data. Related to color parameters and sensors (Equations 13-17).

Table 5. Prediction soil moisture formula with Genetic programming with RGB data (Eq. 1-6), sensors data (Eq. 7-12) as well as RGB and sensors data (Eq. 13-17) as inputs and soil moisture as output

	PREDICTED FORMULA	R^2	MSE	RMSE
1	((-12.235203*(X_3))+(8.179499*(X_2))+28.488047)	0.8354	0.055312029	0.235185095
2	(16.34187*((X_2)*(X_2)))+(-12.452814*(X_3))+29.618999	0.8406	0.053557715	0.231425399
3	11.850106*((X_3)*((X_2)-(X_3)))+25.687898	0.8387	0.054216141	0.232843597
4	(-4.270025*(((X_3)/((X_2)^(X_3)))^(X_3))+28.398337)	0.8434	0.052627867	0.229407644
5	17.473561*(((X_2)^((X_3)^((X_2)*((X_2)-(((X_1)^((X_2)+(X_3)))+((X_2)/(X_1)))/(X_3))))))-(X_3))-5.280389*((X_1)*(X_2))+31.009195	0.8449	0.477804451	0.228297
6	-239.203829*(X_2)+234.831467*((X_2)/(((X_1)^(X_3))*(((X_3)/(X_3))/(((X_3)+(X_1))+(((X_1)*(X_3))-(X_3)))+((X_1)-(((X_1)-(X_1))/((X_1)*(X_1))))))+(X_3)))^(X_2)))-16.207185*(X_3)+32.450448	0.8491	0.474524098	0.22517312
7	6.18095*((X6) - (X5))-0.082796*(X7) + 33.445071	0.6482	1.502913883	2.258750138
8	-0.030255*((((X6) -((X4) * (X5))))) +14.618074	0.6304	0.935107482	0.874426003
9	-1.417192*((G2) +(E2)) +129.063305	0.7884	0.51710931	0.267402039
10	-1.114342 * (x6) -1.476731 * (x4) -0.390532 * (x5) + 120.290384	0.8064	0.505692966	0.255725375
11	-1.219003 * (x6) -1.610459 * (x4) + 121.741966	0.8347	0.486972019	0.237141747
12	-0.911131 * (x5) + -3.095896 * (x4) + -0.192688 * (((((((x4)-((x6)+(x5)))(((((x6)+(x6))+((x6)-(x6)))+(((x5)+(x5))/(x6)))-(x5)))-((((((x6)/(x4))-((x6)(x4)))((x4)/((x4)/(x5))))((((x5)/(x4))(x5))/(x6)))+(x4)))-(x5))/((((((x4)/(x5))((x6)-(x5)))-(x5))/(x5))((((x5)-(x6))(x6))+((x4)(((((x6)+(x6))((x4)-(x5)))-(((x6)(x6))/(x5)))/((x5)/(((x5)(x6))/((x5)+(x5)))))))))*(x4)) + 98.074742	0.8751	0.453245065	0.205431089
13	(6.678679*(X_2))+(-0.029373*((X_5)*((X_3)*(X_4))))+27.506435	0.8668	0.044752655	0.211548233
14	0.476518*(X6)-0.216827*((X5)*((X3)/((X2)^(X1))))+53.947099	0.8865	0.195859392	0.038360901
15	-0.639722*((X5)*(X3))+7.506274*(X2)+28.131956	0.8847	0.197399462	0.038966548
16	-0.392350*(X5)-11.483265*(X3)+7.554770*(X2)+35.160437	0.8837	0.445200195	0.198203213
17	-18.427200*(X2)+155.027205*((((X2)/(X4))/(X5))*(X6))-10.510444*(X3)+27.762460	0.8878	0.441288	0.194735099

X_1= r, X_2= b, X_3= g, X_4= soil temperature, X_5= environment temperature, X_6= humidity, X_7= light

According to the proposed models obtained from the analysis of results in genetic programming the best model with high R2 (R2 = 0.89) and the lowest error (RMSE = 0.038) is a combination model of images and sensors (by comparing the models related to images, sensors and the

combined model of images and sensors) that Figure 4 shows the fit line between the measured data and the original data (Formula 3):

$$S.M= 0.476518*(X6)-0.216827*((X5) *((X3)/((X2) ^(X1)))) +53.947099 \quad (3)$$

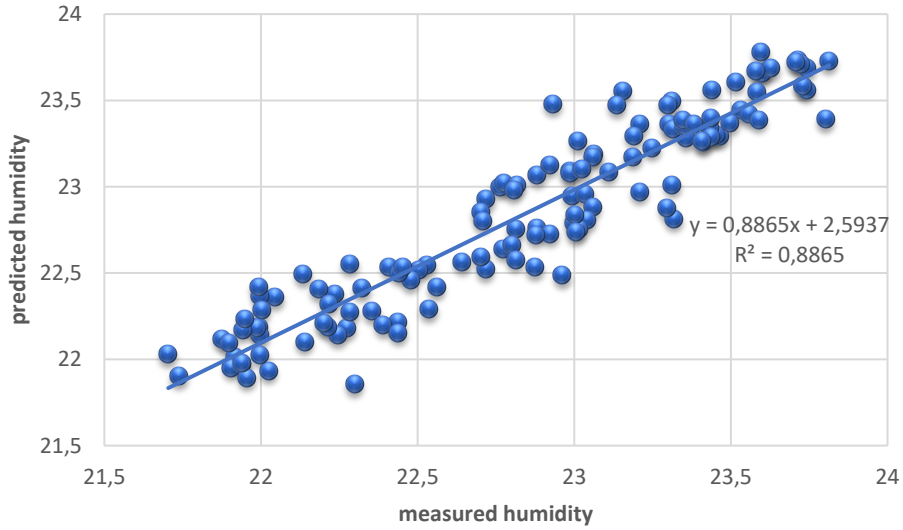


Figure 4. The fit line drawn between actual values and values predicted by genetic programming

In this model, 70% of the data (90 data) were used for train and 30% of the data (38 data) were used for model evaluation (Sanuade et al., 2020). The Fit linear was performed in the training stage between the output (predicted data) and the actual data (measured) (Figures 6 and 5) to evaluate genetic programming, then, the R^2 was calculated for both stages. The two stages of training and testing had a high R^2 equal to 0.802 and 0.806, respectively.

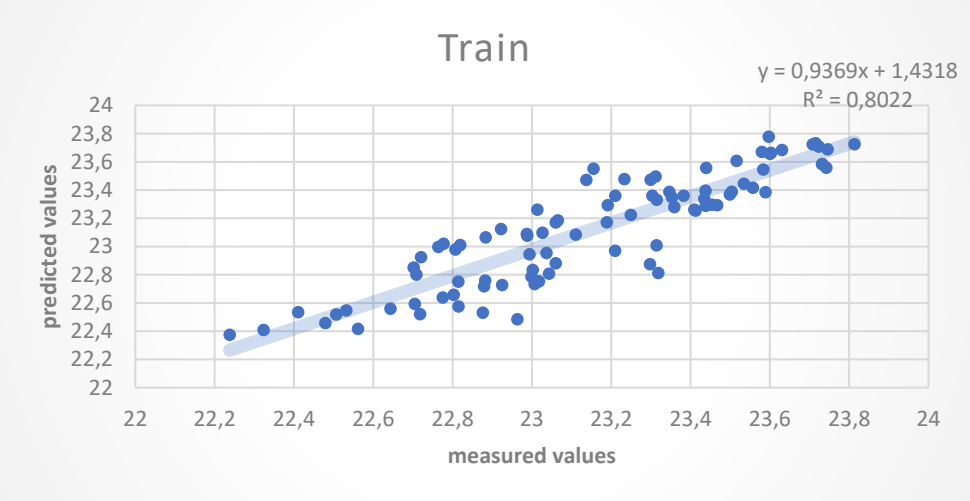


Figure 5. The fit line drawn between actual values and values predicted by genetic programming in the program training stage

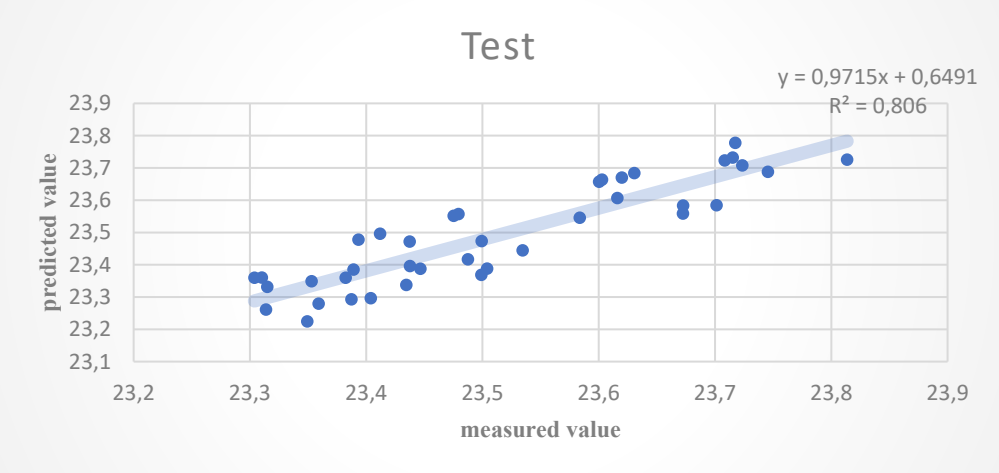


Figure 6. The fit line drawn between actual values and values predicted by genetic programming in the program testing stage

The proposed model from genetic programming (5) is superior compare with the obtained regression model (4). The regression model and genetic programming are as follows:

$$S.M = -0.162\ S.T - 0.310\ r + 0.407\ g - 0.571\ b - 0.145\ en.T - 0.153\ HUM \quad (R^2 = 0.868) \qquad (4)$$

S.M= 0.476518*(X6)-0.216827*((X5) *((X3)/((X2) ^(X1)))) +53.947099 (R^2= 0.8865) (5)

The proposed model to present to the microcontroller for an intelligent irrigation system is a combination model resulting from genetic programming due to the higher R^2 of the genetic programming model (R^2 = 0.8865) and less error (RMSE = 0.038360901) (as a combination of color data and sensors) than the regression model with a lower R^2 (R^2 = 0.868) and higher error standard (0.21). Wen et al. (2022) reported good performance and high accuracy of genetic programming in predicting the optimal model compared to other categories.

4-3. Comparison of yield, water consumption, and color indicators in intelligent irrigation system (next year)

In the second phase of the research (next year), after planning the microcontroller with the optimal model obtained from genetic programming and adjusting the intelligent irrigation system, the crop was planted for the second time and the irrigation system provide the plant's water needs automatically. Therefore, the user only controlled the system to perform the irrigation operation correctly and did not interfere with the steps and time of irrigation. Therefore, water consumption and crop performance between both stages of the research were compared and evaluated to compare and evaluate the performance of the intelligent irrigation system with the previous year (values measured by the user in the first stage).

The yield of the product in the first year (total yield) was equal to 24.40 kg, which had a similar yield in the second year, and therefore no significant difference was observed at the level of 5% between them. As shown in Figure 7, the mean performance of the 3 replicates did not differ significantly. Also, statistical analysis was performed at the level of 5% between the color indices measured in the first and second years, and no significant difference was observed between them. Figures 9, 10, and 11 show the differences between the data measured during the growth period in the first and second stages for the color index r, g, and b, respectively. On the other hand, the amount of water consumption was investigated during the two stages of cultivation.

The amount of water consumption was equal to 136420 ml in the first year when irrigation was done by the user and in the second year, this amount was equal to 121420 ml when irrigation was

done by the intelligent system without user intervention. For this purpose, statistical analysis was performed by T-test.

According to the results, there was a statistically significant difference at the level of 5% between the amount of water consumption in the first and second years. As shown in Figure (8, the amount of water consumption was reduced by the intelligent irrigation system and 11% of the amount of water consumption was saved. Therefore, the farmer can save on an economic scale due to the reduction of water consumption in irrigation and the lack of reduction in crop yield. Also, could save water for larger crops for a long time due to the global water crisis.

Similarly in the research of Biabi et al. (2019), 14,590 ml of water was stored per day for 90 Lilium plants with the development of an intelligent irrigation system. Barkunan et al. (2019) used the intelligence drip irrigation system in rice cultivation and reported that the intelligent irrigation system consumes only 58.57% of water and stores 41.43% of it compared to the traditional flood irrigation system. This is while compared to the drip irrigation system, it consumes 86.97% of water and stores 13.03%. In another study, the wireless sensor network was examined to reduce the consumption of irrigation water in corn the required information from temperature, humidity, wind, and light sensors were received and, Irrigation was ordered to the axes (pipes) according to the water requirement of the product. The efficiency of the system used was estimated at 85% along with the axial drip (pipe) irrigation system. On the other hand, they stated that although water consumption in uncontrolled irrigation was 11% less than in automatic irrigation, due to the automatic control of this system, it is possible to have a higher percentage of optimization and better determine the points that need more irrigation (González-Briones et al., 2018).

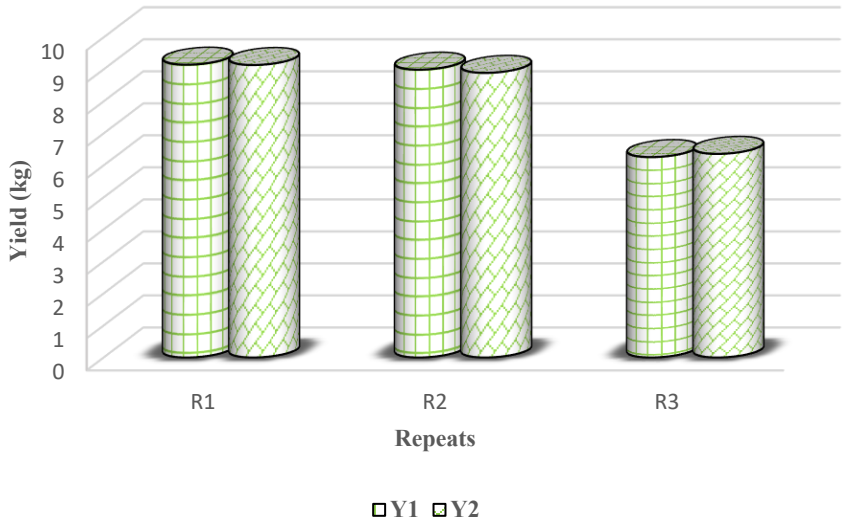


Figure 7. Product yield (kg) in two consecutive years; Y1: irrigation in the first year by the user and Y2: irrigation in the second year by the intelligent system, R1: repeat 1, R2: repeat 2, R3: repeat 3

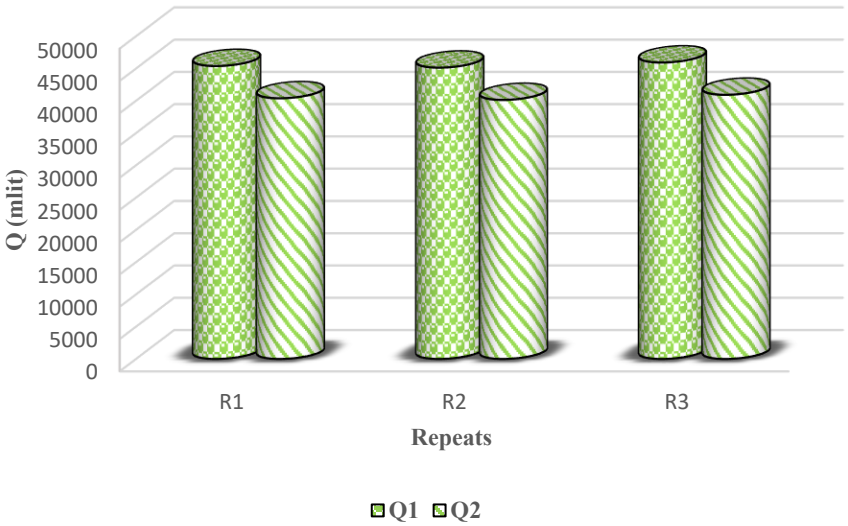


Figure 8. The amount of water used in irrigation (ml) in two consecutive years; Q1: irrigation in the first year by the user and Q2: irrigation in the second year by an intelligent system, R1: repeat 1, R2: repeat 2, R3: repeat 3

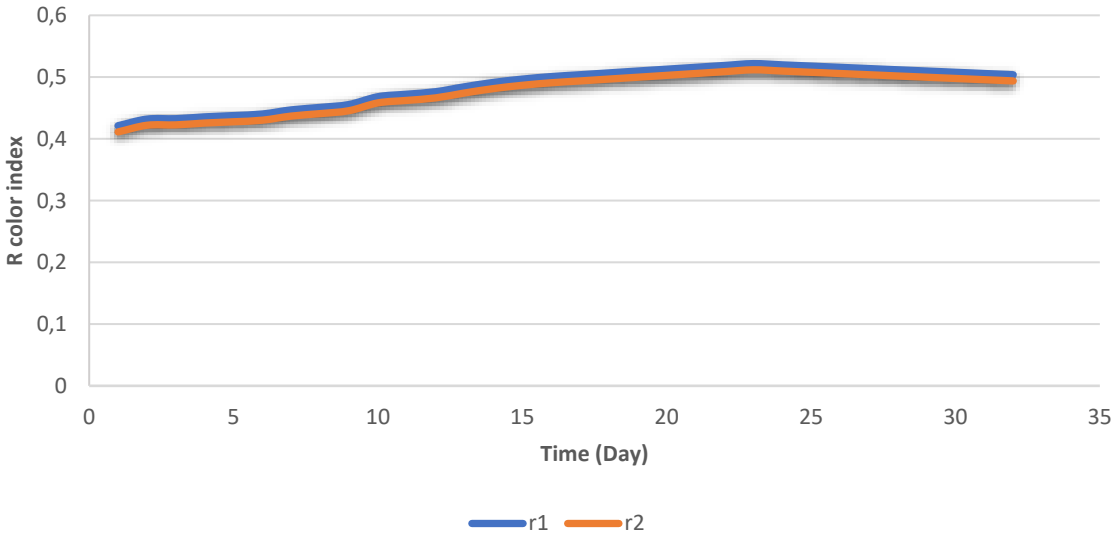


Figure 9. R Color index for two consecutive years; r1: first year and r2: second year

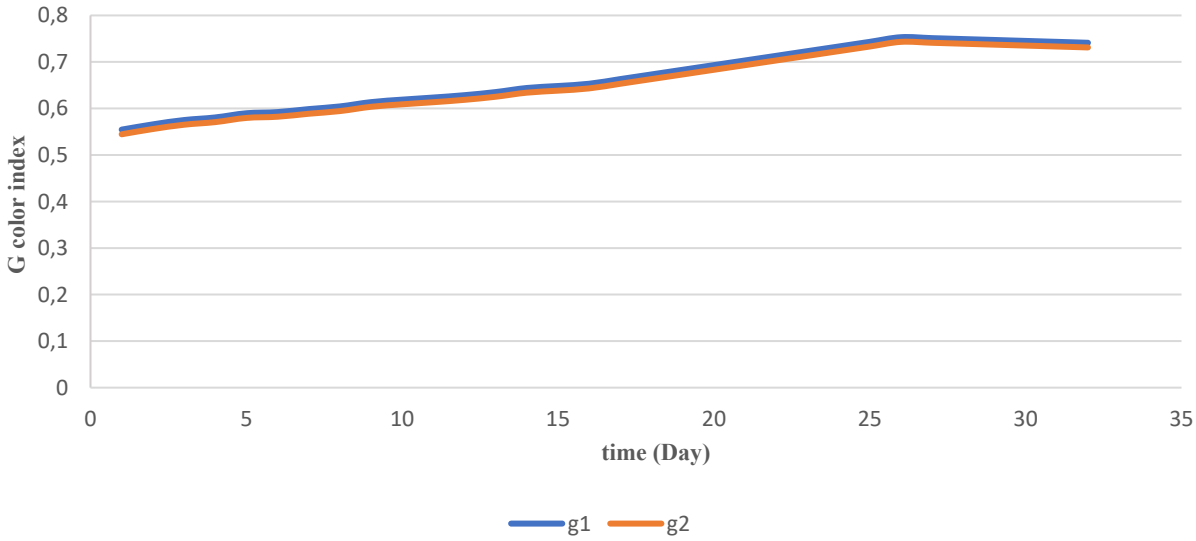


Figure 10. G Color index for two consecutive years; g1: first year and g2: second year

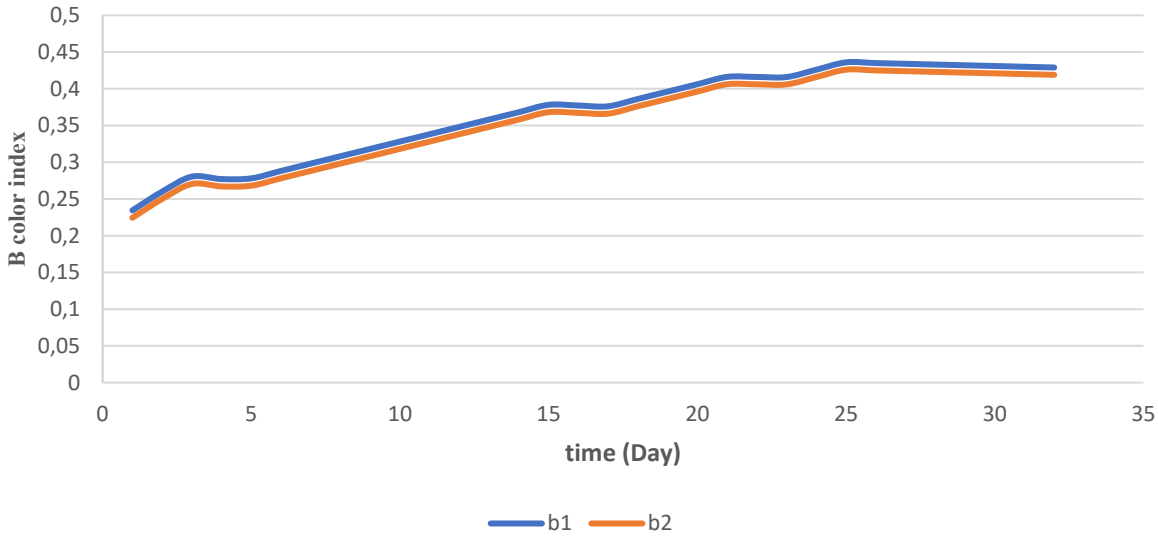


Figure 11. B Color index for two consecutive years; b1: first year and b2: second year

5. Conclusion

In this study, the regression model and genetic programming were evaluated to select and implement an optimal model for the development of an intelligent irrigation system. The irrigation system was programmed based on the optimal model obtained by genetic programming which was a combination of sensor data and RGB color indicators from imaging during the culture period. Also, the irrigation operations were performed automatically in the second period of cultivation and the user received and controlled information about sensors and solenoid valves through mobile phones and the Internet. Changes in yield and color indicators of the crop weren't observed after the second period of cultivation and evaluation of the intelligent irrigation system. But water consumption was reduced and saved by about 11%. Therefore, due to the water shortage crisis in the world and the agricultural sector especially, if this system is used in greenhouse conditions, for a long time, not only it can improve crop yield but also it can save water consumption, and the farmer will suffer less from the perspective of economy.

In future research, intelligent irrigation systems can be developed and planned for other crops. It is also possible to make an adjustment to the system and equip it to detect plant deficiencies (nutritional deficiencies) so it can intelligently decide about the plant's water and nutritional needs.

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